to ensure that the timeout is still valid; `wait_until()` may be more appropriate in many circumstances. The thread may be blocked for longer than the specified duration. Where possible, the elapsed time is determined by a steady clock.

Synchronization

Calls to `notify_one(), notify_all(), wait(), wait_for()`, and `wait_until()` on a single `std::condition_variable_any` instance are serialized. A call to `notify_one()` or `notify_all()` will only wake threads that started waiting *prior* to that call.

Std::Condition_Variable_Any::Wait_For Member Function Overload Th- hat Takes a Predicate

Waits until the `std::condition_variable_any` is woken by a call to `notify_one()` or `notify_all()` and the predicate is `true`, or until the specified time period has elapsed.

Declaration

```
template<typename Lockable,typename Rep,
    typename Period, typename Predicate>
bool wait_for(
    Lockable& lock,
    std::chrono::duration<Rep,Period> const& relative_time,
    Predicate pred);
```

Preconditions

The expression `pred()` shall be valid and shall return a value that's convertible to `bool`. `Lockable` meets the `Lockable` requirements, and `lock` owns a lock.

Effects